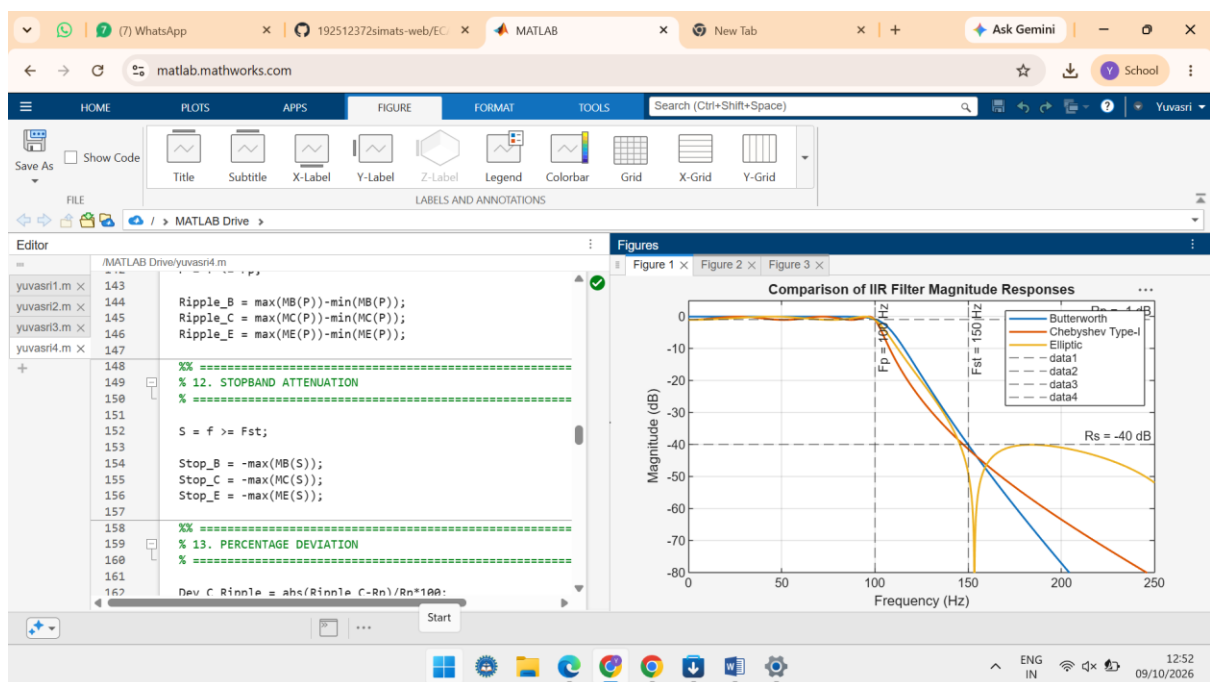
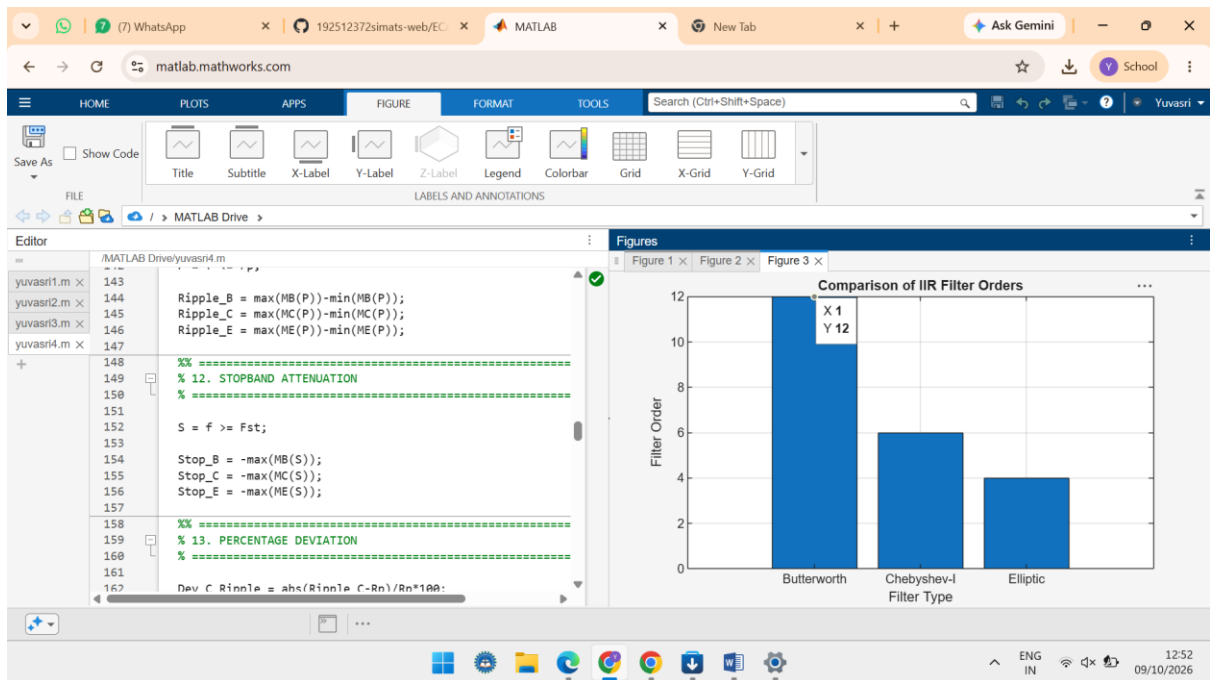
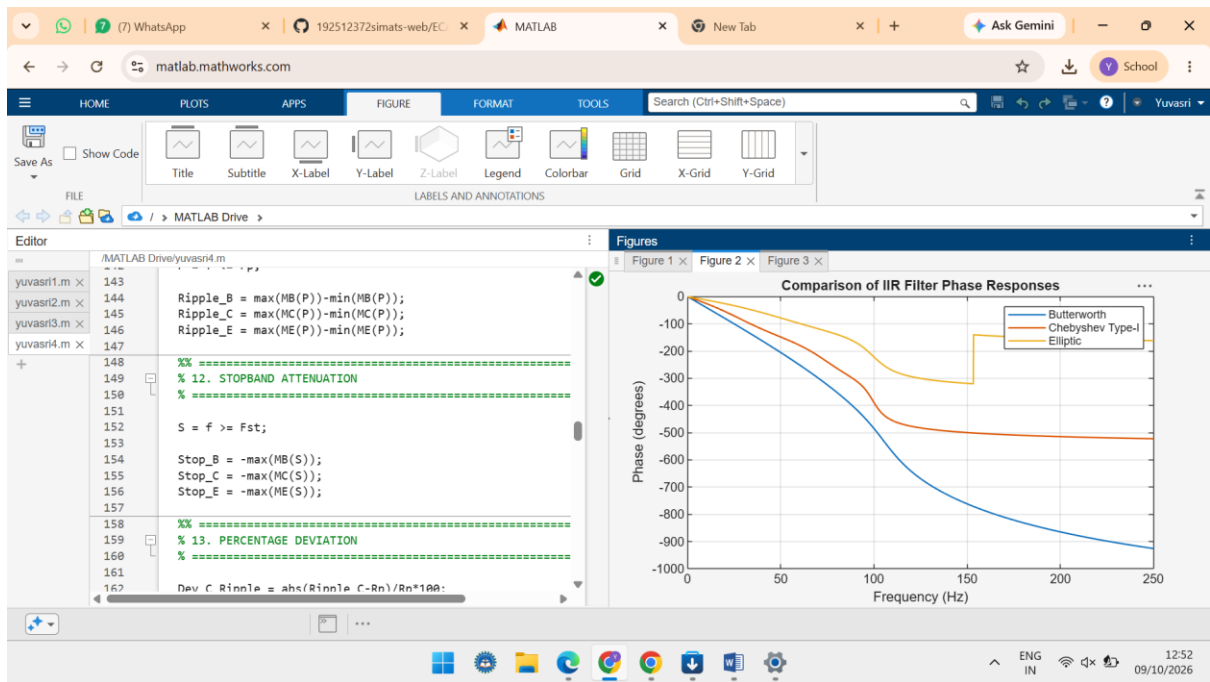


4. Design and Comparative Analysis of Butterworth, Chebyshev and Elliptic IIR Filters Using MATLAB





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MATLAB Drive

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri4.m Figure 1 Figure 2 Figure 3

Command Window

```

=====
IIR FILTER COMPARATIVE ANALYSIS
=====

INPUT PARAMETERS
Sampling Frequency      = 1000.00 Hz
Passband Edge Frequency = 100.00 Hz
Stopband Edge Frequency = 150.00 Hz
Passband Ripple         = 1.00 dB
Stopband Attenuation    = 40.00 dB

CHEBYSHEV TYPE-I FORMULA
Ripple factor epsilon  = 0.5088
Calculated order       = 6.2071
Theoretical minimum N  = 7

=====
FILTER ORDER COMPARISON
=====

```

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MATLAB Drive

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri4.m Figure 1 Figure 2 Figure 3

Command Window

```

=====
FILTER ORDER COMPARISON
=====
Butterworth Order      = 12
Chebyshev Type-I Order = 6
Elliptic Order         = 4

=====
BUTTERWORTH COEFFICIENTS
=====
Numerator b =
1.0e-03 *
0.0002 0.0027 0.0146 0.0488 0.1098 0.1757 0.2050 0.1757 0.1098 0.0488 0.0146 0.0027 0.0002

Denominator a =
1.0000 -6.8844 22.4341 -45.5100 63.7725 -64.8501 48.9643 -27.6092 11.5218 -3.4662 0.7128 -0.0899 0.0053

=====

```

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Command Window

```

Numerator b =
    1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464

Denominator a =
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586

=====
ELLIPTIC COEFFICIENTS
=====
Numerator b =
    0.0197   -0.0171    0.0333   -0.0171    0.0197

Denominator a =
    1.0000   -3.0330    3.8118   -2.2911    0.5554

=====

```

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Command Window

```

PASSBAND RIPPLE
Butterworth      = 0.8033 dB
Chebyshev Type-I = 1.0000 dB
Elliptic         = 1.0000 dB

STOPBAND ATTENUATION
Butterworth      = 40.0198 dB
Chebyshev Type-I = 41.3365 dB
Elliptic         = 40.0000 dB

CHEBYSHEV RIPPLE DEVIATION
Specified Ripple = 1.00 dB
Measured Ripple  = 1.0000 dB
Percentage Deviation = 0.0000 %

TRANSITION CHARACTERISTICS
Passband Edge    = 100.00 Hz
Stopband Edge    = 150.00 Hz
Transition Band   = 50.00 Hz

```

13:17 09/10/2026